

Frequency content of non-periodic signals

- Fourier Series (sines, cosines or complex exponentials of discrete frequencies $\omega_0 k$) are used to analyze the frequency content of periodic signals
- Fourier Transform is used to analyze the frequency content of non-periodic signal.
- Frequency is a continuous parameter.

$$x(t) = \int_{-\infty}^{+\infty} X(f) e^{j2\pi f t} df$$

$$X(f) = \int_{-\infty}^{+\infty} x(t) e^{-j2\pi f t} dt$$

- Fourier Transform
- Discrete time Fourier Transform
- Fast Fourier Transform (for a non-periodic digital signal)
- Time is measured in integers (bins)
- Frequency is measure in bins too.

$$x[n] = \sum_{k=1}^N X[k] e^{j2\pi \frac{kn}{N}}$$

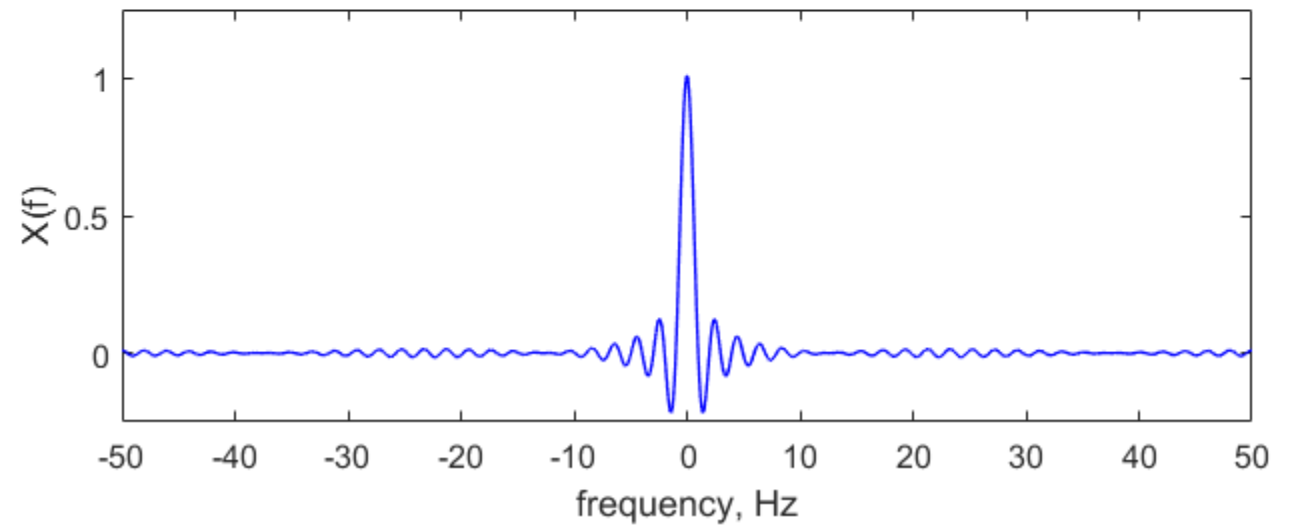
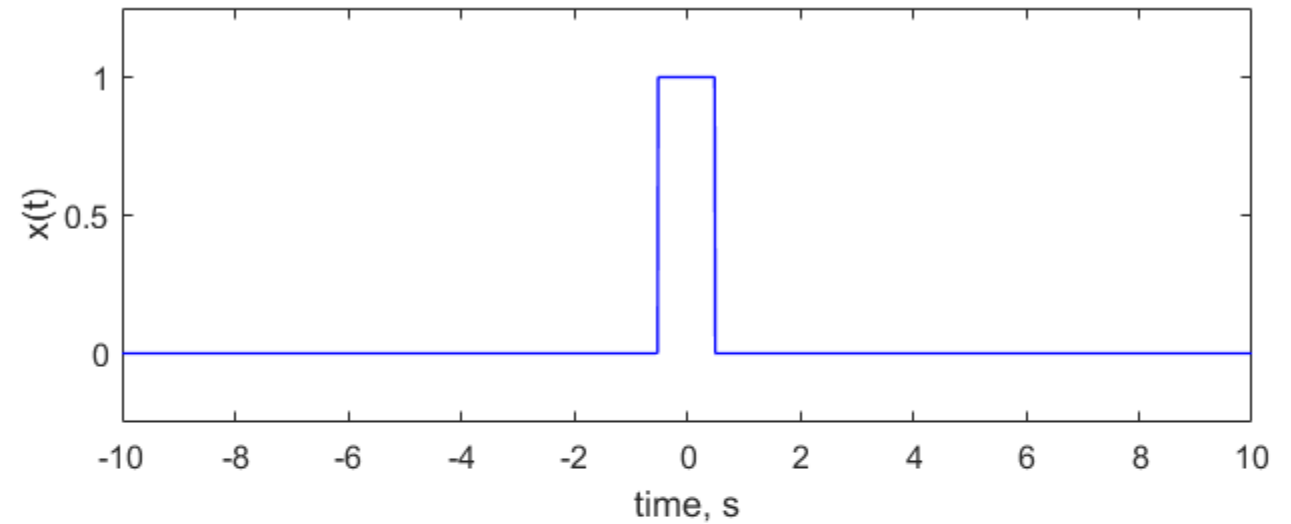
$$X[k] = \frac{1}{N} \sum_{n=1}^N x[n] e^{-j2\pi \frac{kn}{N}}$$

Common Fourier Transform Pairs: Rectangular Pulse

$$\Pi(t) = \begin{cases} 1, & -0.5 < t < 0.5 \\ 0, & \text{otherwise} \end{cases}$$

Fourier transform

$$X(f) = \text{sinc}(f) = \frac{\sin(\pi f)}{\pi f}$$

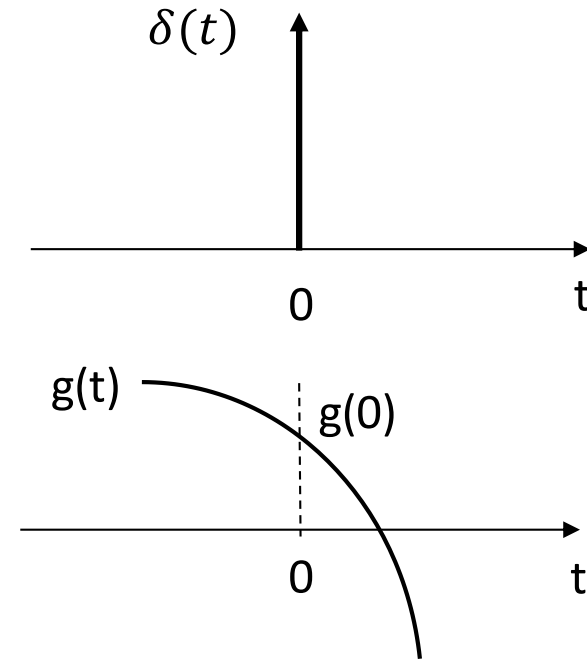


Delta function

- Delta function is a mathematical abstraction.
- It is used to sample other functions.
- Definition:

$$\delta(t) = 0, \text{ for } t \neq 0$$

$$\int_{-\infty}^{+\infty} \delta(t) dt = 1$$



- Fourier transform:

$$\int_{-\infty}^{+\infty} \delta(t) e^{j2\pi t f} dt = 1$$

Time delay

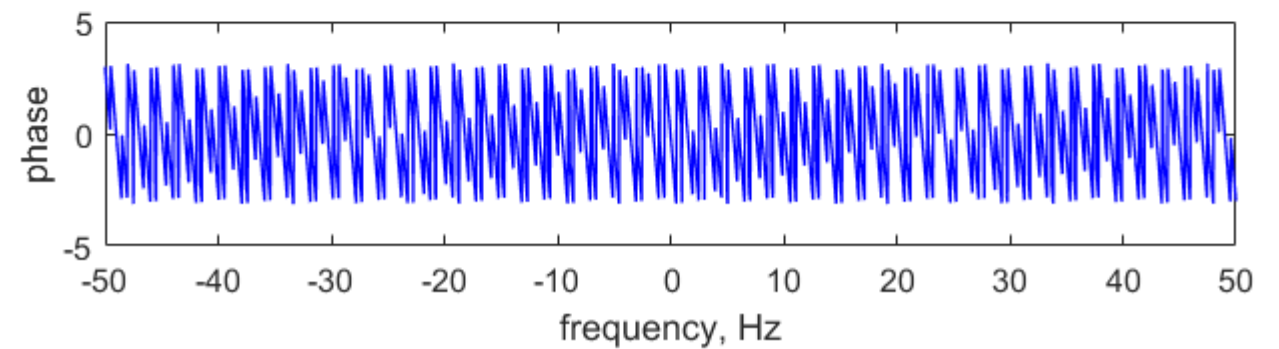
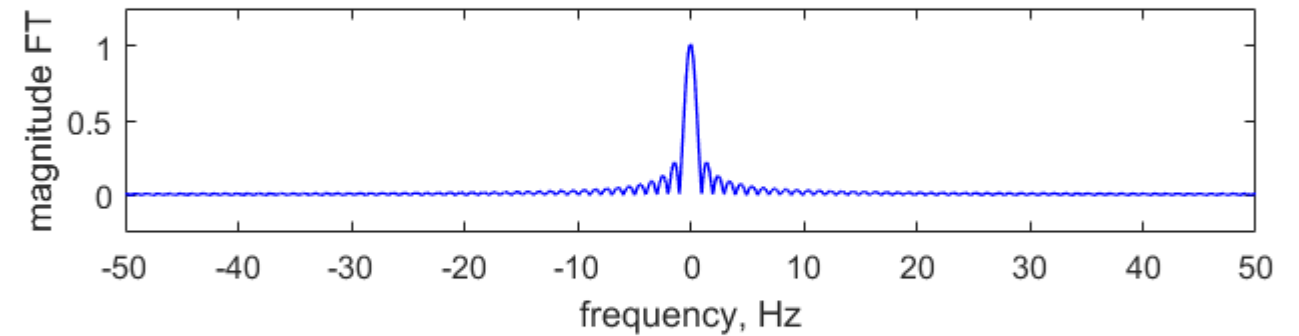
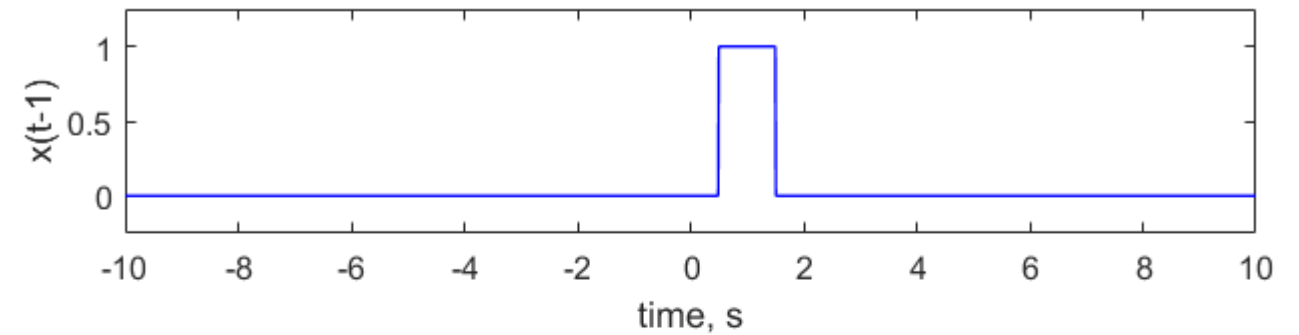
Fourier pairs

$$\Pi(t) \overset{FT}{\Leftrightarrow} \text{sinc}(f)$$

$$\Pi(t - 1) \overset{FT}{\Leftrightarrow} \text{sinc}(f)e^{-j2\pi f}$$

$$\delta(t) \overset{FT}{\Leftrightarrow} 1$$

$$\delta(t - 2) \overset{FT}{\Leftrightarrow} e^{-j2\pi 2f}$$



Linearity of FT

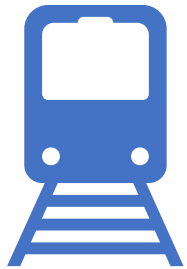
- Fourier transform of a sum of signals = sum of Fourier transforms
- Example

Complex Exponential and Cosine

Fourier pairs (duality of FT and its inverse)

$$e^{j2\pi f_0 t} \stackrel{FT}{\Leftrightarrow} \delta(f - f_0)$$

$$\cos(2\pi f_0 t) = \frac{1}{2} (e^{j2\pi f_0 t} + e^{-j2\pi f_0 t}) \stackrel{FT}{\Leftrightarrow} \frac{1}{2} \delta(f - f_0) + \frac{1}{2} \delta(f + f_0)$$



Train of pulses: FT of
a Dirac train and
another Dirac train,
but in frequency
space.



Correlation

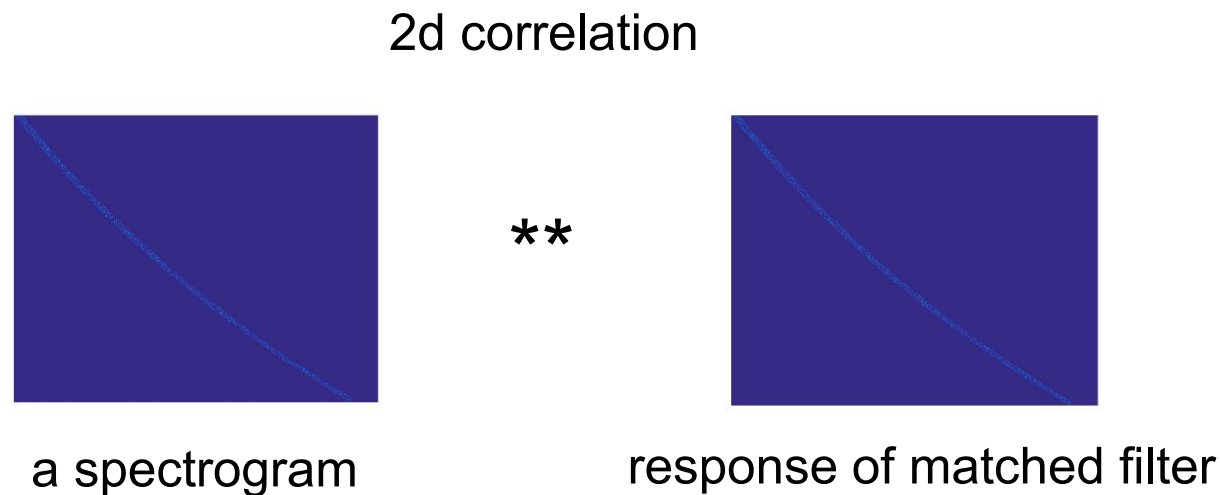
$$g(t) * h(t) \stackrel{FT}{\Leftrightarrow} G(f)H(f)^*$$

“*” stands for complex conjugate

- No shifting, multiplying and integrating at every t
- Only multiplication point-by-point

Matched filtering of filter-bank data

- Correlate filter-bank data with a dispersed pulse.
- Vary pulse width and its shape. Also vary DM value.
- Computationally it is very expensive.
- PSNR can be used as detection criterion.



correlation function



Correlation in 2d FT

- 2d FT stands for two dimensional Fourier Transform

